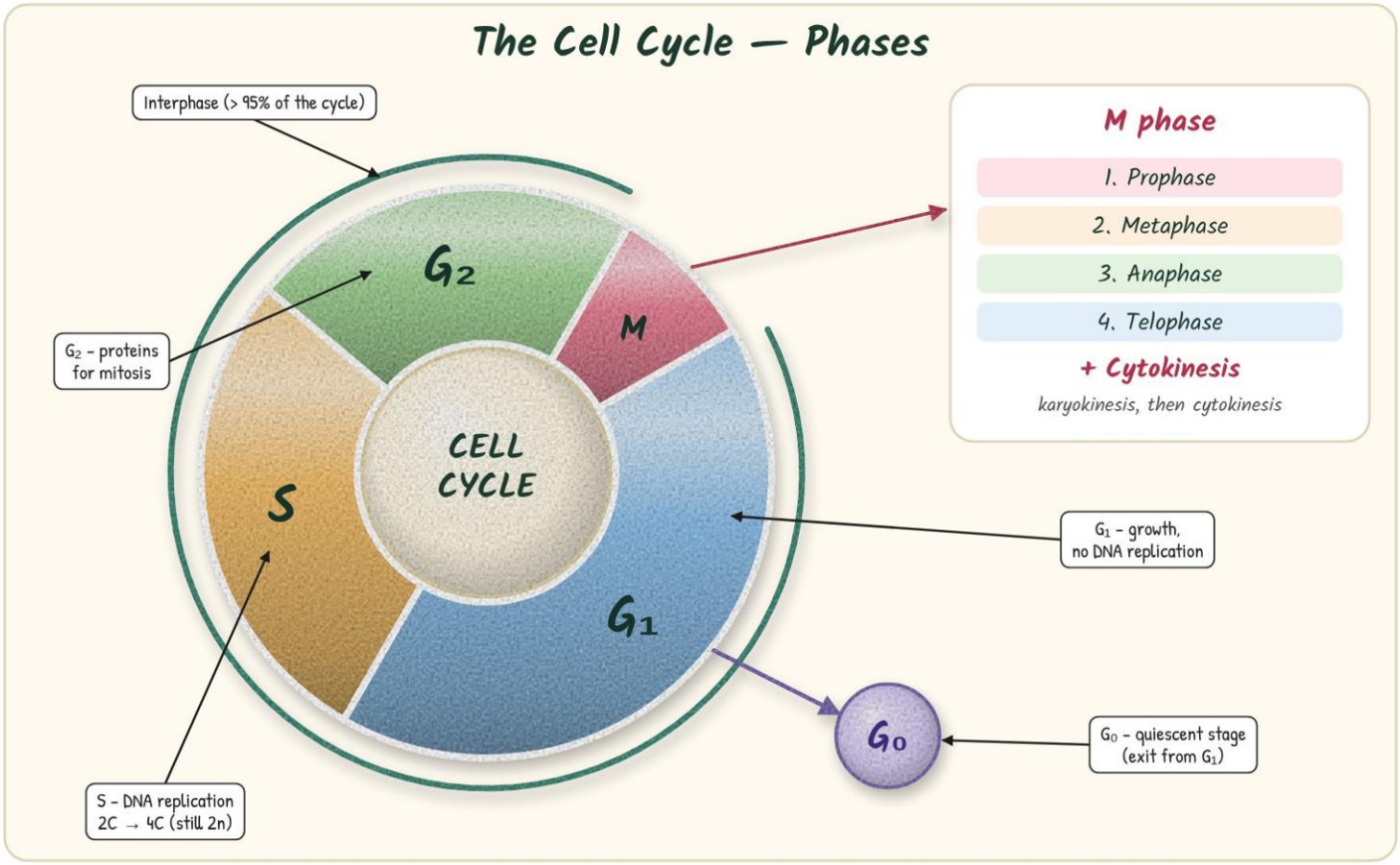
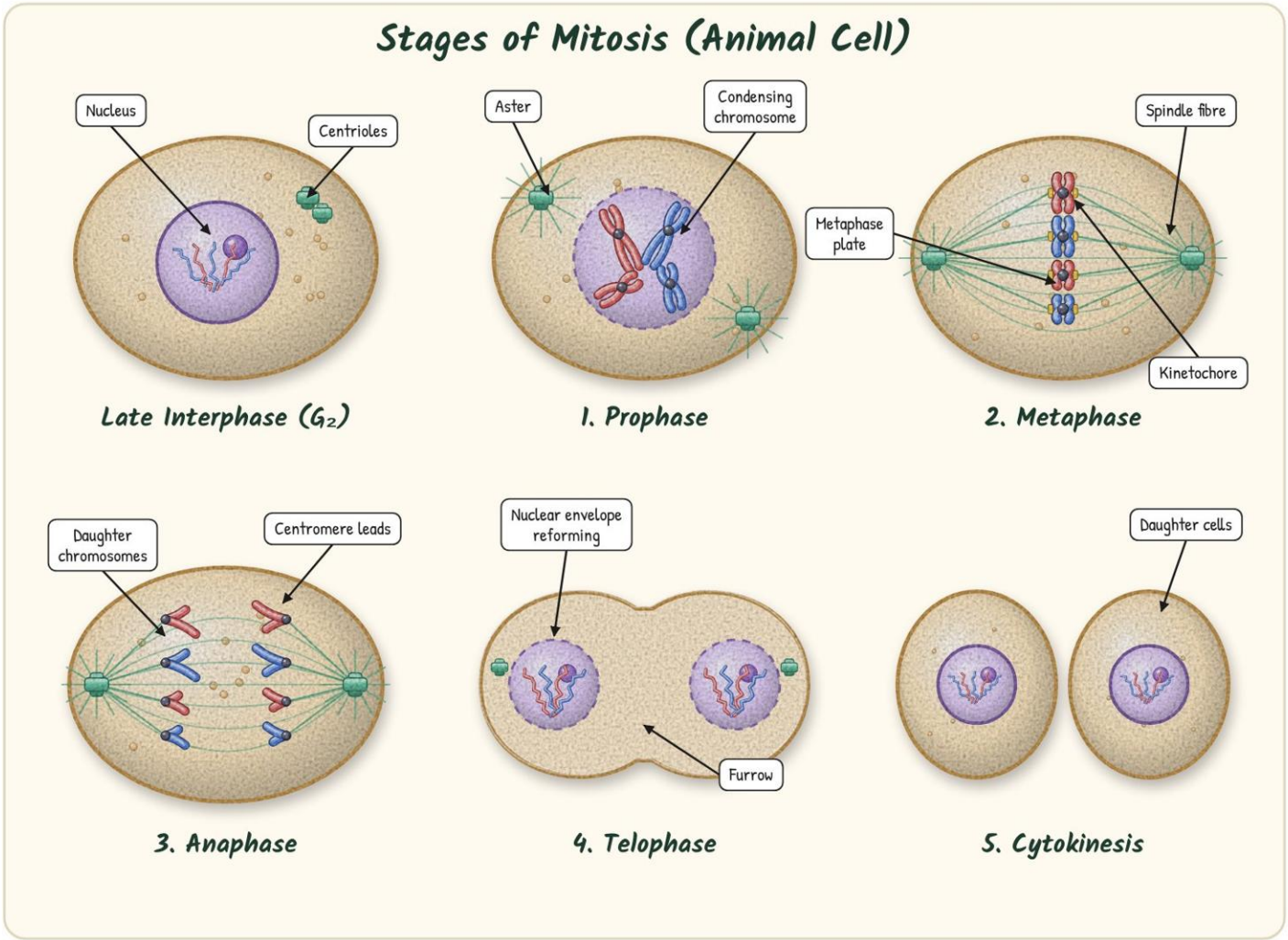
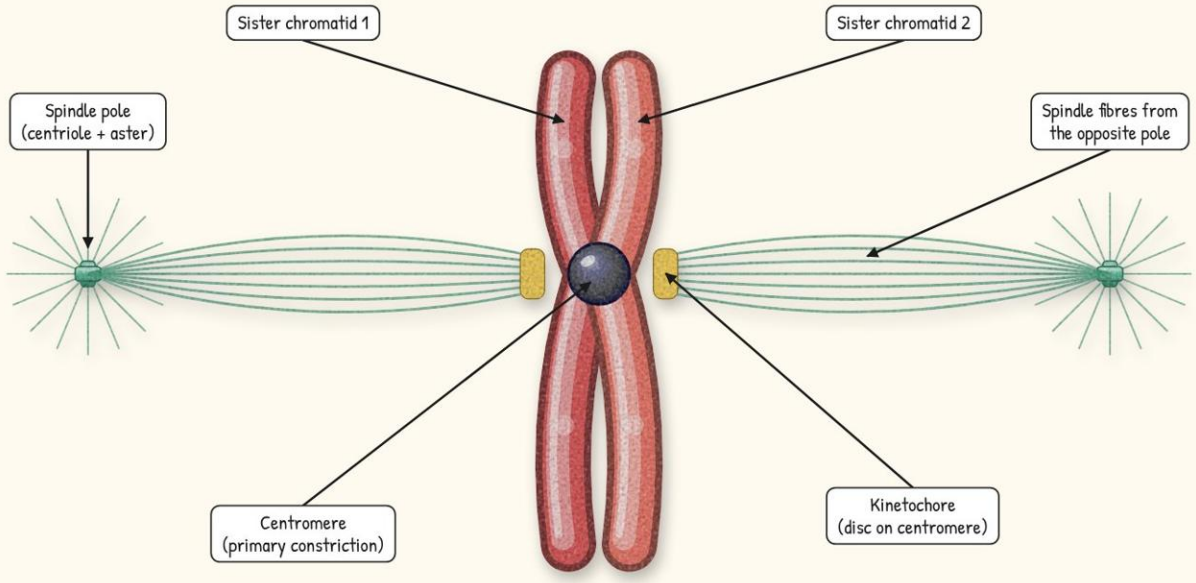

The Cell Cycle — Phases
Interphase (> 95% of the cycle)
G2
M
S
G1
G0
CELL CYCLE
G2 – proteins for mitosis
S – DNA replication 2C → 4C (still 2n)
G1 – growth, no DNA replication
G0 – quiescent stage (exit from G1)
M phase
1. Prophase
2. Metaphase
3. Anaphase
4. Telophase
+ Cytokinesis
karyokinesis, then cytokinesis

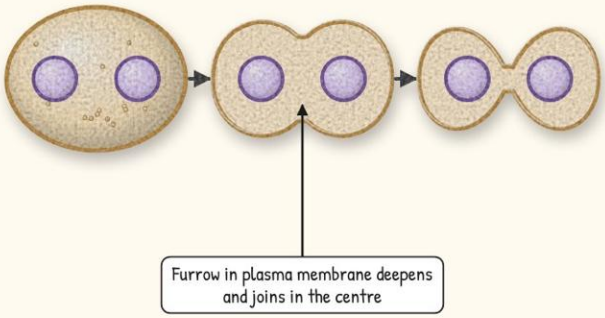

Stages of Mitosis (Animal Cell)
Nucleus
Centrioles
Late Interphase (G2)
Aster
Condensing chromosome
1. Prophase
Spindle fibre
Metaphase plate
Kinetochore
2. Metaphase
Daughter chromosomes
Centromere leads
3. Anaphase
Nuclear envelope reforming
Furrow
4. Telophase
Daughter cells
5. Cytokinesis

Metaphase Chromosome & Kinetochore

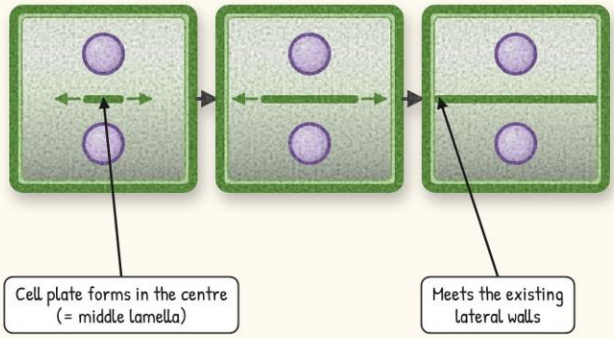


Cytokinesis — Animal Cell vs Plant Cell

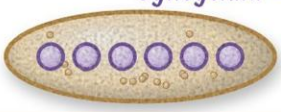
Animal cell: furrowing (centripetal)



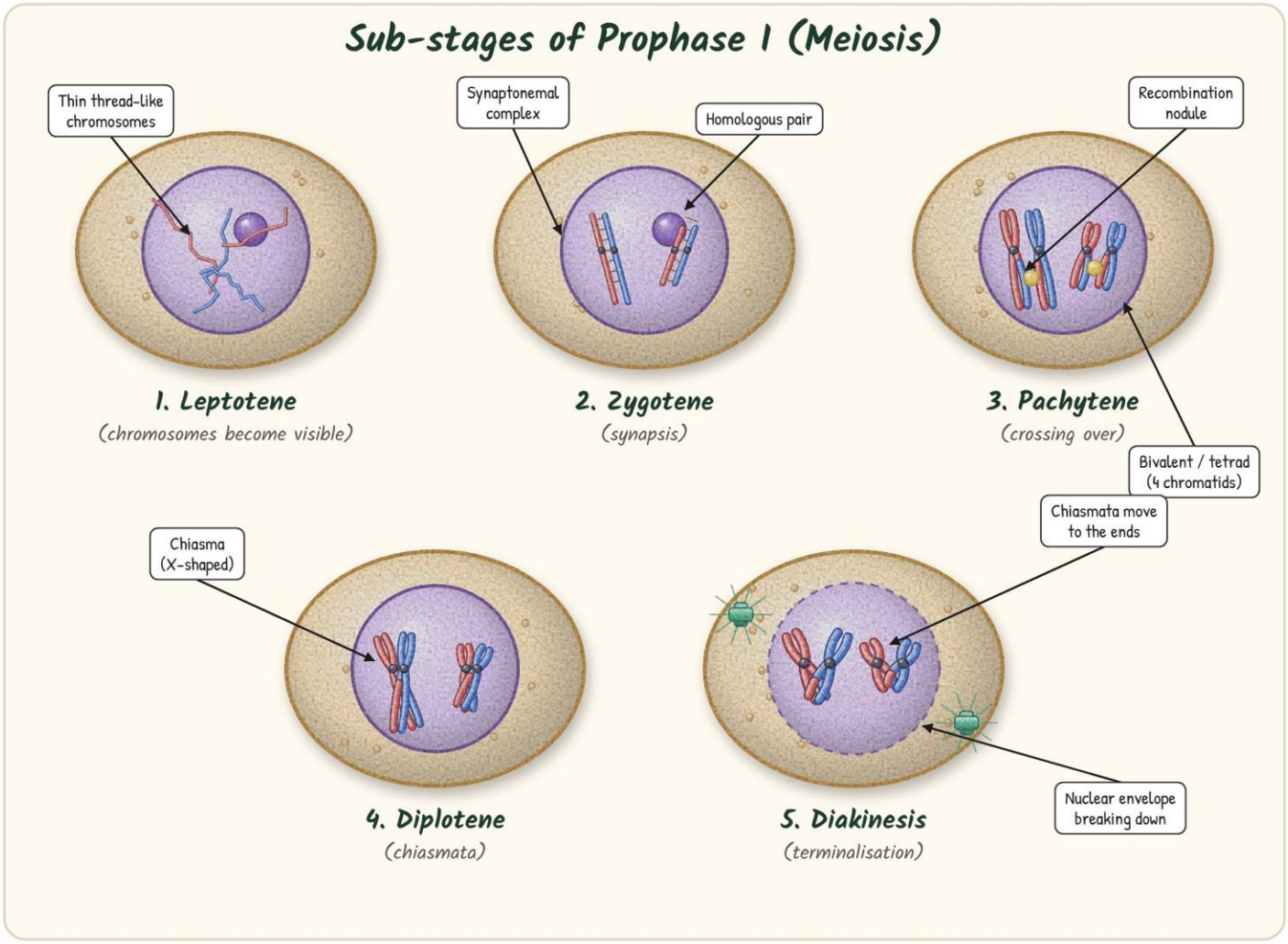
Plant cell: cell plate (centrifugal)

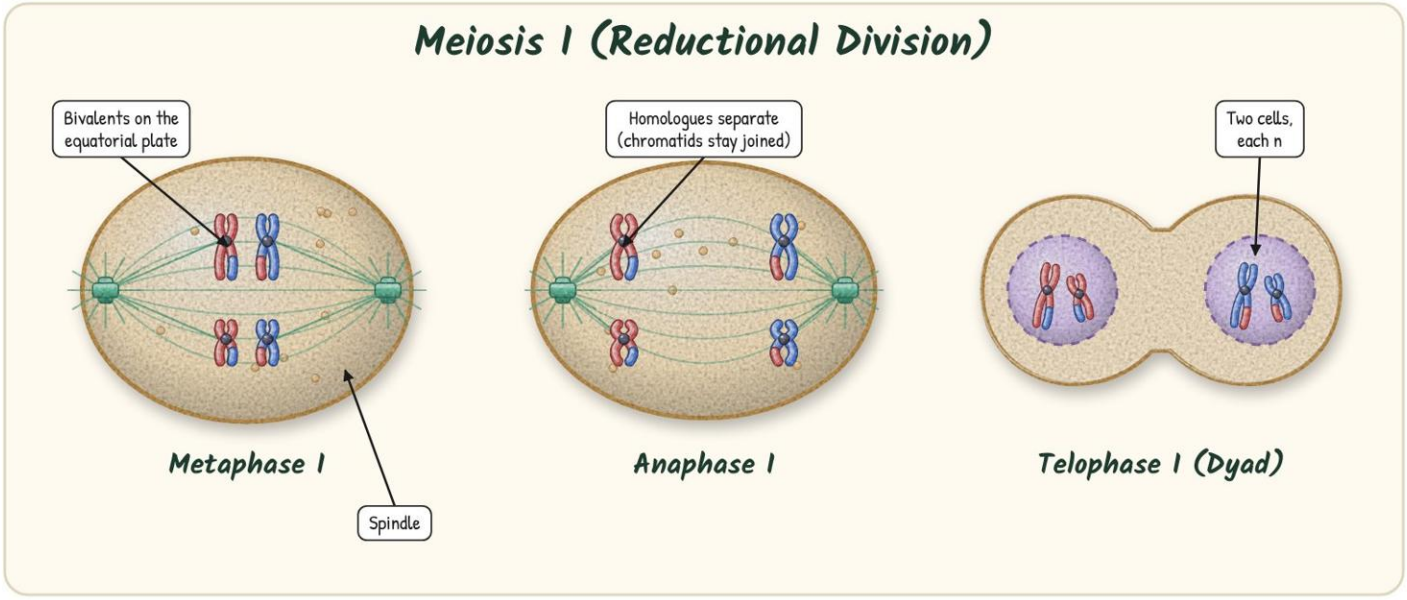


Syncytium — when cytokinesis does not follow karyokinesis

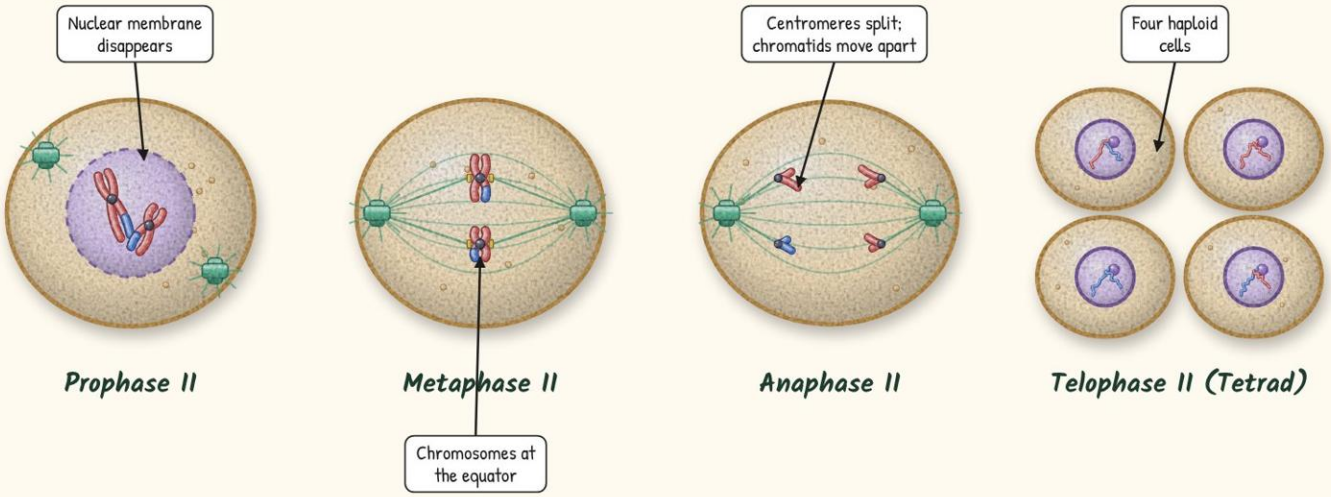


Many free nuclei in one cytoplasm — e.g. liquid endosperm of coconut


Sub-stages of Prophase I (Meiosis)
Thin thread-like chromosomes
1. Leptotene
(chromosomes become visible)
Synaptonemal complex
Homologous pair
2. Zygotene
(synapsis)
Recombination nodule
3. Pachytene
(crossing over)
Bivalent / tetrad (4 chromatids)
Chiasma (X-shaped)
4. Diplotene
(chiasmata)
Chiasmata move to the ends
Nuclear envelope breaking down
5. Diakinesis
(terminalisation)


Meiosis I (Reductional Division)
Bivalents on the equatorial plate
Metaphase I
Spindle
Homologues separate (chromatids stay joined)
Anaphase I
Two cells, each n
Telophase I (Dyad)

Meiosis II (Equational Division)



Mitosis vs Meiosis — the Outcome

